

Manvi Jain

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PROFILE

Physics PhD student and member of the LIGO–Virgo–KAGRA (LVK) Collaboration, working on matched-filter searches for gravitational waves from compact binary coalescences. Current research builds template banks for precessing and tidally deformed binaries, develops waveform models for compact binaries in modified gravity, and uses the analysis pipelines (SGNL/GstLAL) that detect these signals. Co-author of a 2024 *Physical Review Letters* search for asymmetric, heavily precessing binary black holes.

Research interests: Gravitational-wave data analysis; compact-binary coalescence searches; template-bank construction; binary neutron stars with tides and precessing-binary waveforms; tests of general relativity (dynamical Chern–Simons gravity); Fisher-matrix parameter estimation; scientific machine learning; high performance computing; numerical methods.

EDUCATION

Ph.D. in Engineering & Applied Science (Physics), University of Massachusetts Dartmouth 2025 – present

- Advisor: Prof. Sarah Caudill.

M.S. in Physics, University of Massachusetts Dartmouth 2023 – 2025

- Thesis: *Harmonic Decomposition versus Metric-Based Template Banks in Precessing Gravitational-Wave Searches*. Advisor: Prof. Sarah Caudill.

B.Tech. in Electrical & Electronics Engineering, Guru Gobind Singh Indraprastha University (GGSIU), New Delhi 2018 – 2022

RESEARCH EXPERIENCE

Doctoral Researcher, Caudill Gravitational-Wave Group, UMass Dartmouth 2025 – present

- **Tidal template banks for binary neutron star searches.** Constructing matched-filter template banks that incorporate tidal effects for binary neutron star (BNS) systems; running injection campaigns to characterize template-bank recovery, toward a future BNS search. (*manuscript in preparation*)
- **Waveforms for precessing binaries in dynamical Chern–Simons gravity.** Derived analytic equations in Mathematica for precessing compact binaries in dynamical Chern–Simons (dCS) gravity; Fisher-matrix parameter-estimation analysis in progress. (*manuscript in preparation*)

Graduate Research Assistant, UMass Dartmouth / LVK Collaboration 2024 – 2025

- Ran precessing binary black hole searches on LIGO O1 and O2 data with the GstLAL pipeline.
- Packaged a precessing-harmonics waveform-generation code into a reusable module for pipeline use.
- Quantified how weak detector glitches injected onto simulated signals bias signal recovery.
- Contributed the O1/O2 precessing-search analysis underlying Schmidt *et al.*, *Phys. Rev. Lett.* **133**, 201401 (2024).

PUBLICATIONS

Peer-reviewed

1. S. Schmidt, S. Caudill, J. D. E. Creighton, L. Tsukada, A. Doke, **M. Jain**, *et al.*, “Searching for Asymmetric and Heavily Precessing Binary Black Holes in the Gravitational-Wave Data from the LIGO Third Observing Run,” *Physical Review Letters* **133**, 201401 (2024). arXiv:2406.17832; DOI. (author 6 of 42)

In preparation

1. **M. Jain**, S. Caudill, *et al.*, “Tidal template banks for binary neutron star searches” (in preparation, 2026).
2. **M. Jain**, *et al.*, “Parameter estimation for precessing binaries in dynamical Chern–Simons gravity” (in preparation, 2026).

CONFERENCES, SCHOOLS & WORKSHOPS

Gravitational Wave Physics and Astronomy Workshop (Poster presentation), Georgia Institute of Technology, Atlanta GA	2025
SGN Meet, Pennsylvania State University	2024
Women in Data Science (WiDS) Datathon, Microsoft NERD Center, Cambridge MA	2024
“Astrophysics: Cosmology” course, Australian National University	2022
Summer School on Astrophysics & Cosmology, Inter-University Centre for Astronomy & Astrophysics (IUCAA), Pune	2021

TEACHING & MENTORING

Head Teaching Assistant - Electricity & Magnetism Laboratories, UMass Dartmouth	Fall 2026 – present
• Lead TA coordinating lab instruction, staffing, and grading across E&M laboratory sections.	
Teaching Assistant - Electricity & Magnetism I (summer session), UMass Dartmouth	Summer 2025
Head Teaching Assistant - Classical Mechanics Laboratory, UMass Dartmouth	2023 – 2024
• Taught Newtonian mechanics, friction, and momentum conservation using Logger Pro; ran weekly problem-solving tutorials.	
Astronomy Educator & Curriculum Designer - Genex Space, Spark Astronomy, Spaceonova, STEM & Space	2021 – 2023
• Designed beginner-to-advanced astronomy courses and taught 40+ students topics from rocketry and spectroscopy to stellar astrophysics; built laboratory content used by 10+ schools.	
Volunteer Teacher - Pehchaan, The Street School, New Delhi	2019 – 2020

AWARDS & HONORS

Doctoral Fellowship , UMass Dartmouth - Competitive fellowship supporting gravitational-wave research	2025 – 2026
Provisional discovery of two main-belt asteroids (2021 QJ70, 2021 QD44) and 10 preliminary detections, International Astronomical Search Collaboration (IASC)	2021

OUTREACH & SCIENCE COMMUNICATION

- **Organized:** STEM4Girls workshop “Detecting Gravitational Waves,” UMass Dartmouth (2024); stargazing and solar-observation workshops, STEM & Space (2023).
- **Invited talk:** “The Life Cycle of Stars,” *Math and Beyond* podcast (2023).
- **Panel chair:** “Scope of Privatization in the Space Industry,” SSERD (2022).
- **Science communication:** Produced short-form educational videos and written explainers on astrophysics topics like dark matter and dark energy, the origin of cosmic rays, redshift, planetary science for various start-ups -Spark Astronomy, Spaceonova, and Genex Space (2021-2022).
- **Volunteer:** Chaperone, STEM4Girls, UMass Dartmouth (2023).

TECHNICAL SKILLS

- **Programming:** Python (NumPy, SciPy, Matplotlib, Pytorch, sklearn), Mathematica, C, Julia, Bash.
- **Gravitational-wave related libraries:** GstLAL, Bilby, PyCBC, LALSuite.
- **Methods:** matched filtering, template-bank construction, software injections, Bayesian and Fisher-matrix parameter estimation, normalizing flows / scientific machine learning.
- **Tools & computing:** Git/GitHub, L^AT_EX, Linux, high-performance computing - Slurm, HTCondor.

SELECTED COURSEWORK & TRAINING

- **Graduate coursework:** Scientific Machine Learning; High Performance Scientific Computing; Numerical Analysis; General Relativity; Gravitational Waves; Quantum Field Theory.
- **Graduate course projects:** Massive-graviton alternative theory of gravity; gravothermal catastrophe and systems with negative specific heat; field-theoretic description of gravity (review); 2D conformal field theory.
- **Training:** Responsible Conduct of Research, UMass Dartmouth (2023).